

Unit - II

Data Arrangements and Blood Relations:

- **Linear Arrangement** :- Linear arrangement is the arrangement of given items with the help of the given clues. The main idea here is to represent the given data in an organized manner. Once that is done, solving individual problems becomes very easy.

For example :- Five friends A, B, C, D and E are sitting in a bench s.t.

- E is second to the right of B.
- B is in the middle position.
- A sits at one extreme end.
- C is the neighbour of A.

Find their seating arrangement.

A C B D E

- Read the following information and answer the questions below:
Six swimmers A, B, C, D, E and F compete in a race.
The outcome is as follows:
 - Exactly 2 swimmers finish ahead of B.
 - Only two swimmers separate E and D.

- A is behind D and E.
- B is ahead of E, with one swimmer in-between.
- F is ahead of D.

(i) Who stood fifth in the race?

- (a) A (b) B (c) C (d) D (e) E

(ii) How many swimmers separate A and F?

- (a) 1 (b) 2 (c) 3 (d) 4 (e) Can't determined.

- Directions for [Question No. 1 to 4]

A, B, C, D, E, F and G are sitting in a row facing North:

① - F is to the immediate right to E.

- E is 4th to the right of G.

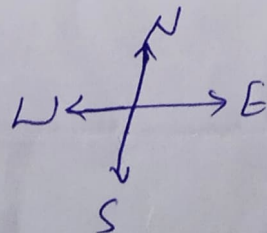
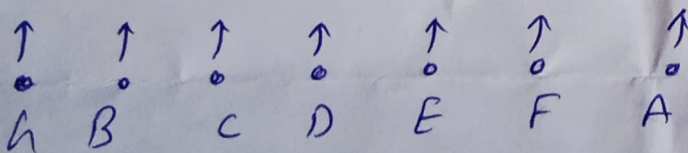
- C is the neighbour of B and D.

- Person who is third to the left of D is at one of ends.

1. What is the position of A?

- (a) Between E and D (b) Extreme left
(c) Centre (d) Extreme right.

Answer- D



A is in extreme right.

2. Who are the neighbours of B?

- (a) C and D
- (b) C and G
- (c) G and F
- (d) C and E

Ans B. (i.e. C & G are the neighbours of B.)

3. Which of the following statement is not true?

- ~~A~~ ~~E~~ ~~is~~ (a) E is to the immediate left of D.
(b) A is at one of the ends.
(c) G is to the immediate left of B.
(d) F is second to the right of D.

Answer A.

4. Who are to the left of C?

- (a) Only B
- (b) G, B and D
- (c) G and B
- (d) D, E, F and A.

Answer C.

Circular Arrangement

Questions on circular arrangements involve arrangement of people sitting around a table or arrangement of things in a circular manner.

In case of people sitting around a table, the table could be of any shape - rectangular, square, circular etc.

Circular arrangement is different from linear arrangements in that there is no first or last position and hence there is no reference point. Therefore, the positions taken by the objects are only relative to each other.

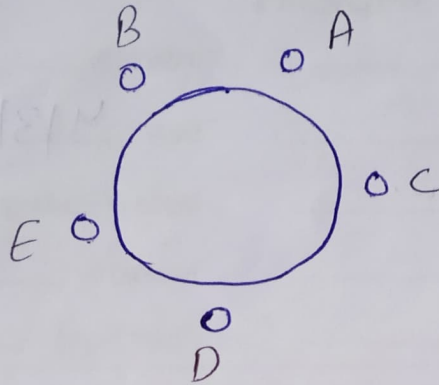
Once you read the question, first draw the shape specified in the question and then draw the slots for the arrangement. Then solve the given clues one by one.

Example:- A, B, C, D and E are sitting in a round table conference and the seating is according to the following conditions:-

- A always sits second to the right of D.
- B always sits to the right of A.
- C and A are neighbours.

Find the ~~or~~ order in which they are seated.

Solⁿ



Directions for [Question No-1 to 5]

Study the following information and answer the questions given below.

Eight people E, F, G, H, J, K, L and M are sitting around a circular table facing the centre. Each of them is of a different profession Chartered Accountant, Columnist, Doctor, Engineer, Financial Analyst, Lawyer, Professor and Scientist but not necessarily in the same order. F is sitting second to the left of K. The Scientist is an immediate neighbour of K. There are only ~~the~~ three people between the Scientist and E. Only one person sits between the Engineer and E. The Columnist is to the immediate right of the Engineer. M is second to the right of K. H is the Scientist. G and J are immediate neighbours of each other. Neither G nor J is an Engineer. The Financial Analyst Analyst is to the

immediate left of F. The ~~Lawyer~~ Lawyer is second to the right of the Columnist. The professor is an immediate neighbour of the Engineer. His second to the right of the chartered Accountant.

1. Which of the following statements is true according to the given arrangement?

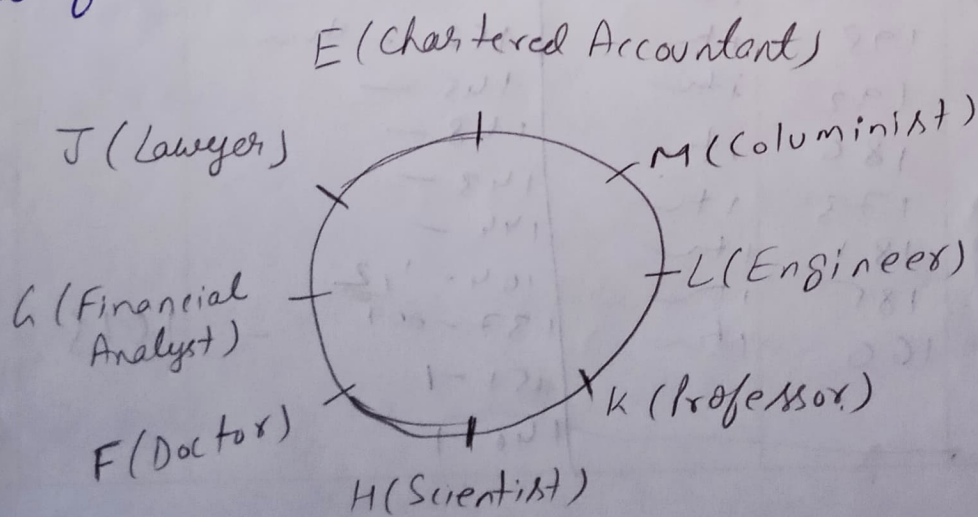
(a). The Lawyer is second to the left of the Doctor.

(b) E is an immediate neighbour of the Financial Analyst.

(c) H sits exactly between F and the Financial Analyst.

(d) Only four people sit between the Columnist and F.

(e) All of the given statements are true.



Ans (a).

2. What is the position of L with respect to the Scientist?

(a) Third to the left (b) Second to the right

(c) Second to the left (d) Third to the right

(e) Immediate right.

Ans (b)

3. Four of the following five are alike in a certain way based on the given arrangement and hence form a group. Which of the following does not belong to that group?

(a) Chartered Accountant - H

(b) M - Doctor

(c) J - Engineer

(d) Financial Analyst - L

(e) Lawyer - K.

Ans C.

4. Who amongst the following is the Professor?

(a) F

(b) L

(c) M

(d) K

(e) J.

Ans D.

5. Who is sitting second to the right of E?

(a) The Lawyer

(b) G

(c) The Engineer

(d) F

(e) K.

Ans b.

Multi-dimensional Arrangement

- In this type of problems, we have to match two or more variables. These questions typically contain clues to club one object (variable) to another in either one to one mapping or one to many mapping.

Example:- Each of the five persons - Naveen, Bala, Raju, Arun and Gopal is the owner of a field in a different city among A through E. Each of them planted different kinds of plants among - guava, mango, apple, banana, and watermelon.

The following are known about them:

- Raju owns a field in city A and he has not planted mango or watermelon.
 - Watermelon grown in a farm in city C, which is owned by Bala.
 - City E grows either apple or banana.
 - Naveen has a field in city D but he is not growing mangoes.
 - Banana or guava ~~also~~ does not grow in city A.
1. If Arun planted mango, then who owns a field in city D?
(a) Raju (b) Arun (c) Naveen (d) Bala (e) None of these.

Naveen	Bala	Raju	Arun	Gopal
D	C	A	E / B	B / E
Guava	↓ Watermelon	↓ Apple	banana / Mango	Mango / banana

2. Who owns a field in city A?

- (a) Raju (b) Arun (c) Bala (d) Naveen
 (e) Cannot be determined.

Ques. Each of the four persons A, B, C and D wear a different coloured shirt - red, pink, blue and white. A has a red shirt and D does not have a pink shirt. C wears a white shirt. Match the shirt colours with the right persons.

Solⁿ

A	B	C	D
Red	Pink.	white	Blue

Ques. Amit, Bharati, Anupama, Deepak and Meenakshi are five friends sitting in a restaurant. They are wearing caps of five different colours Yellow, Blue, Green, white, and red. Also, they are eating five different snacks - burgers, sandwiches, ice-cream, pastries and pizza.

Given data-

- The person wearing a red cap is eating pastries.
- Amit does not eat ice-cream, and Anupama is eating sandwiches
- Bharati is wearing a yellow cap and Amit wearing a blue cap.
- Meenakshi is eating pizza and is not wearing a green cap.

Give the Arrangement:

Solⁿ

Person	Amit	Bharati	Anupama	Deepak	Meenakshi
Colour	Blue	Yellow	Green	Red	White
Dish	Burger	Ice Cream	Sandwiches	Pastries	Pizza.

Blood Relation

① Pointing to a photograph of a boy Suresh said, "He is the son of the only son of my mother." How is Suresh related to that boy?

- (a) Brother (b) Uncle (c) Cousin (d) Father.

Ans (d)

Exp. The boy in the photograph is the only son of Suresh's mother i.e. the son of Suresh. Hence, Suresh is the father of boy.

② If $A+B$ means A is the mother of B; $A-B$ means A is the brother of B; $A\%B$ means A is the father of B and $A\times B$ means A is the sister of B, which of the following shows that P is the maternal uncle of Q?

(a) $Q-N+M\times P$

(b) $P+S\times N-Q$

(c) $P-M+N\times Q$

(d) $Q-S\%P$

Ans C.

Exp. $P-M \rightarrow P$ is the brother of M.

$M+N \rightarrow M$ is the mother of N.

$N\times Q \rightarrow N$ is the sister of Q.

Therefore, P is the maternal uncle of Q.

- ③ If A is the brother of B; B is the sister of C;
and C is the father of D, how D is related to A?
(a) Brother (b) Sister (c) Nephew (d) Cannot be determined.

Ans D

Exp: As the sex of D is not known, hence the relation between D and A cannot be determined.

[If D is male, the answer is Nephew
If D is Female, the answer is Niece]

- ④ If $A+B$ means A is the brother of B; $A-B$ means A is the sister of B and $A \times B$ means A is the father of B.
Which of the following means that C is the son of M?

(a) $M-N \times C+F$

(b) $F-C+N \times M$

(c) $N+M-F \times C$

(d) $M \times N-C+F$

Ans D.

Exp: $M \times N$ = M is the father of N

$N-C$ = N is the sister of C

and $C+F$ = C is the brother of F.

Hence, M is the father of C or C is the son of M.

- ⑤ Introducing a boy, a girl said, "He is the son of the daughter of the father of my Uncle." How is the boy

Exp:- NBP $\Rightarrow$ N is the brother of P.

P-L $\rightarrow$ P is the sister of L.

L+E $\rightarrow$ L is the mother of E.

E-M $\rightarrow$ E is the sister of M.

Hence, L is the mother of M, P is the maternal aunt of M
and N is the maternal uncle of M.

⑧ Amit said - "This girl is the wife of the grandson of my mother". How is Amit related to the girl?

(a) Brother (b) Grandfather (c) Husband.

(d) Father in law.

Ans (d)

The girl is the wife of grandson of Amit's mother i.e.
the girl is the wife of son of Amit, Amit is the
father in law of the girl.

~~⑨ A and B are children of D. Who~~

⑨ Pointing towards a man, a woman said, "His mother is the only daughter of my mother." How is the woman related to the man?

(a) Mother (b) Grand mother (c) Sister (d) Daughter.

Ans (a)

Only daughter of my mother $\rightarrow$ myself.

Hence, the woman is the mother of the man.

related to the girl. ?

- (a) Brother (b) Nephew (c) Uncle (d) Son in law.

Ans ~~(A)~~ (a).

The father of the boy's Uncle $\rightarrow$ the ~~gran~~ grandfather of the boy and daughter of the grandfather $\rightarrow$ sister of father.

Directions to Solve:-

Each of these questions (6-7) is based on the following information:-

① 1. $A+B$ means A is the mother of B.

2. $A-B$ means A is the sister of B.

3. $A \times B$ means A is the father of B.

4. $A \beta B$ means A is the brother of B.

⑥ Which of the following means Q is the grandfather of P?

(a) $P+N \times M \times Q$

(b) $Q \times N \times M + P$

(c) $Q \beta M \beta N \times P$

(d) None of these.

Ans D

~~⑦~~ ⑦ Which of the following means that N is the maternal Uncle of M?

(a) $N \beta P - L + E - M$

(b) $N - Y + A \beta M$

(c) $M - Y \times P - N$

(d) $N \beta C + F \times M$.

Ans - ~~(A)~~ (a)